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15.1 Introduction

Severe acute kidney injury (AKI) occurs in more than 5 % of the critically ill patients and has been associated with mortality rates ranging from approximately 40 % to more than 70 % [1–6]. In the absence of specific pharmacologic interventions, the management of AKI is primarily supportive including optimization of volume status, treatment of electrolyte and acid–base disturbances, nutritional support, and adjustment in medication dosing. Renal replacement therapy (RRT) is often required as an adjunctive therapy, particularly when more conservative management is unable to prevent progressive volume overload and normalize electrolyte and acid–base disturbances or when AKI is prolonged and progressive azotemia or uremic manifestations supervene. Multiple modalities of RRT may be used in the management of critically ill patients with AKI including conventional intermittent hemodialysis (IHD), prolonged intermittent renal replacement therapy (PIRRT; also known as extended duration dialysis, EDD or sustained low-efficiency dialysis, SLED), peritoneal dialysis, and various modalities of continuous renal replacement therapy (CRRT). The modalities of CRRT provide extracorporeal blood purification with an intent to deliver uninterrupted therapy for 24 h a day. CRRT facilitates slow fluid removal and provides greater ability to maintain fluid balance in patients who require large daily fluid intake and provides more stable control of electrolyte and other solutes as compared to conventional IHD.

15.2 History

Peter Kramer and colleagues published the initial description of continuous arteriovenous hemofiltration (CAVH) for the management of patients with oliguric AKI and volume overload in 1977 [7]. Their technique utilized an extracorporeal circuit with a hemofilter interposed between the femoral artery and vein to provide ultrafiltration (UF) and maintain biochemical balance [7–9]. Subsequently, Paganini and colleagues described a similar method for the performance of slow continuous ultrafiltration (SCUF) for the management of volume overload in oliguric patients [10]. Further description of clinical experience and operational characteristics by others including Luer et al. in 1983 [11], Kaplan et al. in 1984 [12], and Golper in 1985 [13] heralded wider adoption of CAVH. Relatively low rates of solute clearance, particularly in patients with high catabolic states, was a major limitation to CAVH and led to the use of counter-current dialysate flow to augment the clearance of urea and other small solutes in continuous arteriovenous hemodialysis (CAVHD) and continuous arteriovenous hemodiafiltration (CAVHDF) [14, 15]. However, these techniques could not overcome the disadvantages inherent to an arteriovenous (AV) circuit including dependence of blood flow on the pressure gradient between mean arterial pressure and central venous pressure, which limited flow, particularly in hypotensive and hemodynamically unstable patients, and complications related to prolonged arterial cannulation, including bleeding, thromboembolism, and infection. These inherent problems with the AV circuit led to the development of pumped venovenous (VV) systems [16–18]. The use of a VV circuit, however, sacrificed the technical simplicity of the AV system, necessitating the addition of air detectors and pressure monitors. While early VV circuits were cobbled together from individual components, progressive improvement in technology has yielded sophisticated machines with integrated blood and fluid pumps, pressure monitoring alarms, blood leak detectors, and UF controls to assure accurate fluid balance. With progressive advances in technology, CRRT has advanced

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from a treatment used by a small number of innovators for the most desperately ill patients to a mainstream therapy that is widely used for the management of critically ill patients with AKI,

15.3 Modalities of Continuous Renal Replacement Therapy

CRRT is defined as any extracorporeal blood purification-technique intended to substitute for impaired kidney function that is designed to be utilized on a continuous (i.e., 24 h per day) basis. These modalities may be provided using either an AV or a VV extracorporeal circuit. In AV therapies, blood enters the extracorporeal circuit through a cannula in an artery, flows through the hemofilter/hemodialyzer driven by the pressure gradient between mean arterial pressure and central venous pressure, and is returned through a catheter in the femoral or central veins. In the VV therapies, vascular access is provided by a double-lumen catheter in the femoral or central veins, with pump-driven blood flow through the extracorporeal circuit. While the use of VV modalities require greater technical complexity, including blood pumps, pressure monitors, and return-line air detectors, the ability to provide higher blood flow rates independent of mean arterial blood pressure and avoidance of prolonged arterial cannulation has led to predominant use of these modalities [19].

Beginning in the mid-1990s, a standard terminology was developed to describe the multiple modalities of CRRT (Table 15.1) [20–22]. SCUF provides UF for the management of volume overload (Fig. 15.1). Since the UF rate does not exceed the refilling of the intravascular compartment from the interstitium, intravenous replacement fluids are not required. Although SCUF is effective for the management of volume overload, for example, in patients with decompensated heart failure, solute clearance rates are negligible and it is not suitable for the management of azotemia or the acid-base and electrolyte disturbances associated with kidney failure. In continuous hemofiltration (continuous arteriovenous hemofiltration, CAVH; continuous venovenous hemofiltration, CVVH), high-volume UF is performed at rates exceeding those needed for volume management and the excess ultrafiltrate is replaced intravenously with a balanced electrolyte solution (Fig. 15.2). Solute removal is primarily convective, with clearance rates of urea and other

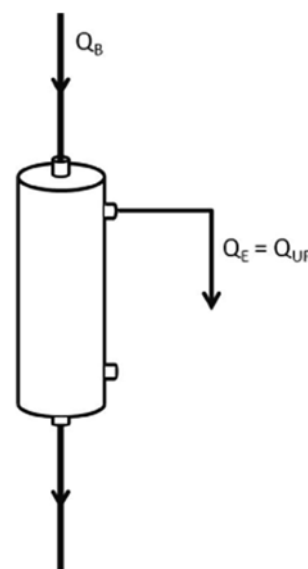


Fig. 15.1 Slow continuous ultrafiltration. Schematic diagram illustrating slow continuous ultrafiltration. Blood is shown flowing through a hollow-fiber hemofilter. An ultrafiltrate is generated at a rate equal to the rate of desired fluid loss and is discarded as effluent (Q_B blood flow, Q_E effluent flow, Q_{UF} ultrafiltrate flow)

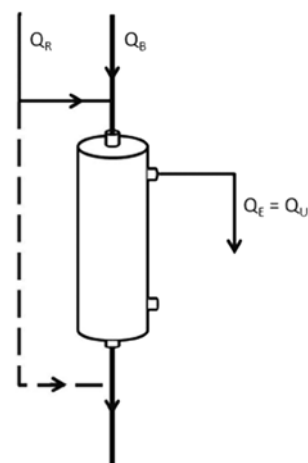


Fig. 15.2 Continuous hemofiltration. Schematic diagram illustrating continuous hemofiltration. Blood is shown flowing through a hollow-fiber hemofilter. An ultrafiltrate is generated at a rate which exceeds the desired rate of net fluid loss. Intravenous fluid is administered either prior to the hemofilter (*solid line*) or after the hemofilter (*dashed line*) to replace the volume of ultrafiltrate that exceeds the desired net fluid loss. The effluent volume is equal to the ultrafiltration rate (Q_B blood flow, Q_E effluent flow, Q_{UF} ultrafiltrate flow, Q_R replacement fluid)

Table 15.1 Modalities of continuous renal replacement therapy

Modality	Volume management	Solute clearance	Replacement fluids	Dialysate
Slow continuous ultrafiltration (SCUF)	Yes	Minimal	No	No
Continuous hemofiltration (CAVH, CVVH)	Yes	Yes	Yes	No
Continuous hemodialysis (CAVHD, CVVHD)	Yes	Yes	No	Yes
Continuous hemodiafiltration (CAVHDF, CVVHDF)	Yes	Yes	Yes	Yes

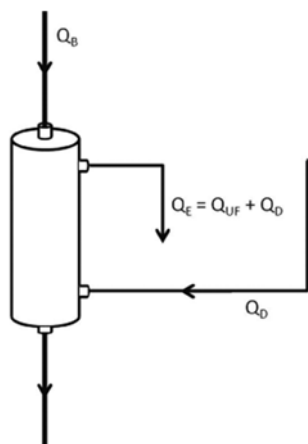


Fig. 15.3 Continuous hemodialysis. Schematic diagram illustrating continuous hemofiltration. Blood is shown flowing through a hollow-fiber hemodialyzer. Dialysate is perfused through the ultrafiltrate compartment countercurrent to the direction of blood flow. Ultrafiltration occurs at the rate desired for net negative fluid balance. The effluent volume is equal to the rate of dialysate flow plus the ultrafiltration rate (Q_B blood flow, Q_E effluent flow, Q_{UF} ultrafiltrate flow, Q_D dialysate)

small solutes proportional to the total UF rate. In continuous hemodialysis (continuous arteriovenous hemodialysis, CAVHD; continuous venovenous hemodialysis, CVVHD), dialysate is perfused across the opposite side of the membrane from the blood and solutes are removed primarily by diffusion down a concentration gradient (Fig. 15.3). As in SCUF, UF is limited to the rate necessary for the optimization of volume status and intravenous replacement fluids are not required. Continuous hemodiafiltration (continuous arteriovenous hemodiafiltration, CAVHDF; continuous venovenous hemodiafiltration, CVVHDF) combines the convective solute removal of continuous hemofiltration with the diffusive solute removal of continuous hemodialysis (Fig. 15.4). Dialysate is perfused through the hemodiafilter on the opposite side of the membrane from the blood, permitting diffusion of low-molecular-weight solutes while high-volume UF is performed to provide convective solute removal, with the ultrafiltrate volume exceeding the desired rate of volume removal replaced intravenously with a balanced electrolyte solution.

15.4 Mechanisms of Solute Removal During CRRT

RRT involves the transfer of fluid and solutes across a semipermeable membrane. Although fluid and solute flux is predominantly from the blood into the dialysate/ultrafiltrate compartment, diffusion and backfiltration from the dialysate/ultrafiltrate compartment may occur. In the modalities of CRRT, UF of fluid is driven by hydrostatic pressure across

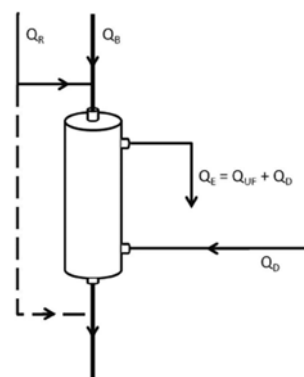


Fig. 15.4 Continuous hemodiafiltration. Schematic diagram illustrating continuous hemodiafiltration. Blood is shown flowing through a hollow-fiber hemodiafilter. Dialysate is perfused through the ultrafiltrate compartment countercurrent to the direction of blood flow. An ultrafiltrate is generated at a rate which exceeds the desired rate of net fluid loss. Intravenous fluid is administered either prior to the hemofilter (*solid line*) or after the hemofilter (*dashed line*) to replace the volume of ultrafiltrate that exceeds the desired net fluid loss. The effluent volume is equal to the rate of dialysate flow plus the ultrafiltration rate (Q_B blood flow, Q_E effluent flow, Q_{UF} ultrafiltrate flow, Q_R replacement fluid, Q_D dialysate)

the hemodialysis/hemofilter membrane and solute removal occurs via diffusion, convection, or a combination.

15.4.1 Ultrafiltration

Ultrafiltration (UF) describes the removal of plasma water across a semipermeable membrane driven by a transmembrane pressure gradient (TMP). The ultrafiltration rate (Q_{UF}) is determined by the product of the hydraulic permeability of the membrane (K_M), the membrane surface area (A), and the transmembrane pressure gradient (TMP).

$$Q_{UF} = K_M \times A \times \text{TMP}.$$

The transmembrane pressure is the difference between the driving effect of hydrostatic pressure (ΔP) and restraining effect from the oncotic pressure (Π) generated by plasma proteins (since the ultrafiltrate is essentially protein free, the oncotic pressure of the ultrafiltrate is considered to be zero).

$$\text{TMP} = \Delta P - \Pi$$

In the AV modalities of CRRT, blood flow rates and the transmembrane hydrostatic pressure were relatively low. As UF occurred over the length of the hemofilter, the oncotic pressure in the blood compartment increased, providing a significant restraining effect on UF, occasionally to the point where $\Pi \approx \Delta P$ and further UF ceased. In the pump-driven VV modalities of CRRT, the blood flow rate and ΔP are sufficiently

high that oncotic pressure is of little consequence as a determinant of UF.

15.4.2 Diffusion

Diffusion is the movement of solutes in solution from a region of higher concentration to a region of lower concentration. During dialytic therapies, diffusive clearance occurs when there is a concentration gradient between the blood and dialysate across the semipermeable dialysis membrane. The diffusive flux of a given solute (J_D) is a function of membrane characteristics including surface area (A) and thickness (Δx), the diffusion coefficient of solute (K_D), and the solute concentration between the blood and dialysate compartments (ΔC), and can be expressed as:

$$J_D = K_D \times A \times \left(\frac{\Delta C}{\Delta x} \right).$$

In order to maximize the concentration gradient between the blood and dialysate compartments, dialysate is commonly run countercurrent to the direction of blood flow on the opposite side of the membrane. As solutes diffuse from blood into dialysate, the solute concentration in the blood decreases while the corresponding concentration in the dialysate increases. By maintaining countercurrent flow, the concentration gradient is maximized along the length of the dialysis membrane. The diffusivity of a given solute is inversely related to its molecular weight. Thus, diffusive therapies, such as continuous hemodialysis (CHD), are more efficient in clearing low-molecular-weight solutes (<500 Da), such as urea and creatinine, as compared to higher molecular weight solutes (>1500 Da), such as β_2 -microglobulin and cytokines.

15.4.3 Convection

Convection is the bulk flow of solute across a semipermeable membrane as the result of solvent flux across the membrane. During UF, solutes are entrained in the flow of plasma water across the dialysis membrane, a process referred to as “solvent drag.” Solute flux is therefore dependent upon the net flux of solvent across the membrane (ultrafiltration rate, Q_{UF}) and is independent of the concentration gradient across the membrane. The ability of a given solute to cross a semipermeable membrane by convection is a function of the charge and molecular radius of the solute and the structure of the pores in the membrane. Thus, in contrast to diffusion, the relative clearance of lower- and higher molecular weight solutes are similar; with restriction in convective flux only as the solute’s molecular radius approaches the effective size of the pores in the membrane.

The ability of a solute to cross a membrane by convection is quantified by the membrane reflection coefficient, σ , with values ranging from 0, for solutes that are not restricted by the membrane, to 1, for solutes that are unable to cross the membrane. In clinical practice, however, it is more common to express the ability of solutes to cross the membrane by convection in terms of the sieving coefficient (S), which is equal to $1 - \sigma$ and can be readily calculated from the ratio of solute concentration in the ultrafiltrate (C_{UF}) and blood (C_B):

$$S = \frac{C_{UF}}{C_B}.$$

The membrane’s sieving coefficient for a specific solute may decline over the duration of treatment. Plasma proteins, which are too large to cross the membrane, accumulate along the membrane surface, a phenomenon known as protein concentration polarization. This accumulation of proteins along the membrane surface may restrict access to membrane pores and alter the charge characteristics of the membrane surface [23].

The convective flux (J_C) of a solute is expressed as the product of the ultrafiltration rate (Q_{UF}), the concentration of the solute in the blood (C_B) and the membrane’s sieving coefficient for the solute (S):

$$J_C = Q_{UF} \times C_B \times S.$$

Since convective clearance (K_C) is equal to convective solute flux divided by the solute concentration in blood (J_C/C_B), convective clearance of a solute can be expressed as the product of the UF rate and the sieving coefficient:

$$K_C = Q_{UF} \times S.$$

For low-molecular-weight solutes such as urea, S is approximately unity. Thus, convective clearance for these solutes approximates the UF rate.

Theoretically, the biophysical mechanisms of solute transport in each of the different modalities of CRRT should result in different solute clearance profiles. Continuous hemofiltration, a predominantly convective therapy, should provide more efficient removal of higher molecular weight species than continuous hemodialysis, a predominantly diffusive therapy. The clinical relevance of this is, however, uncertain, as clinical outcomes with hemofiltration and hemodialysis are similar [24]. Filtration and backfiltration occurring over the length of the dialysis membrane during continuous hemodialysis, particularly when very high-flux membranes are used, can result in significant unmeasured convection within the hemodialyzer, augmenting the amount of convective clearance during the predominantly diffusive therapy [25].

15.5 Calculation of Solute Clearance

Solute clearance during extracorporeal RRT can be calculated based on either the disappearance of solute from the blood or from its appearance in the effluent. Since during continuous hemodialysis, the dialysate flow rate is usually substantially less than the blood flow rate and during hemofiltration the solute concentration in the ultrafiltrate is similar to that in plasma water, the change in solute concentration over the length of the hemodialyzer/hemofilter tends to be small. As a result, solute clearance (K) during CRRT is most commonly calculated based on solute appearance in the hemofilter/hemodialyzer effluent [26], expressed as:

$$K = \frac{[(Q_E \times C_E) - (Q_D \times C_D)]}{C_B},$$

where Q_E and Q_D are the effluent outflow and dialysate inflow rates, respectively, and C_E , C_D , and C_B are the solute concentrations in effluent, dialysate, and blood, respectively. Since the ultrafiltration rate (Q_{UF}) is equal to the difference between the effluent outflow rate and dialysate inflow rate ($Q_{UF} = Q_E - Q_D$), the calculation of solute clearance may be rewritten as:

$$K = \frac{Q_{UF} \times C_E}{C_B} + \frac{Q_D \times (C_E - C_D)}{C_B}.$$

Since, when there is no dialysate flow, $Q_E = Q_{UF}$ and C_E/C_B is equal to the sieving coefficient (S), the first term of this equation may be rewritten as $Q_{UF} \times S$, which represents the convective component of solute clearance (K_C). The second term of the equation, $Q_D \times (C_E - C_D)/C_B$, represents the solute clearance that would occur in the absence of UF and can be considered to represent the diffusive component of clearance (neglecting any convective flux resulting from filtration/backfiltration over the length of the hemofilter/hemodialyzer). For solutes that are not present in dialysate ($C_D = 0$), this term simplifies to $Q_D \times C_E/C_B$, where C_E/C_B represents the degree of solute equilibration between blood and dialysate. Thus, similar to convective clearance, diffusive clearance can be considered to be approximated by the product of the dialysate flow rate times and fractional equilibration of solute between blood and dialysate

15.5.1 Determinants of Convective Clearance

As previously discussed, solute clearance in continuous hemofiltration is equal to the product of the UF rate and sieving coefficient. For solutes with relatively low molecular weights, such as urea, sieving coefficients are approximately unity and solute clearance is equal to the UF or effluent flow rate [27, 28]. For higher molecular weight solutes, a more complex relationship may be observed. For example, the

sieving coefficient for β -2 microglobulin is less than one but increases with increasing UF rate, resulting in a nonlinear increase in clearance with increasing effluent flow [27, 28]. The mechanism for this phenomenon is incompletely understood. Higher transmembrane pressure required to augment UF rates may result in recruitment of additional pores and/or changes in pore geometry allowing for the increase in sieving coefficient. Solute concentration polarization may also contribute to the augmentation of convective flux.

Blood flow rate and membrane surface area are of secondary importance in determining convective clearance. Increasing UF at a constant blood flow rate increases the fraction of plasma water removed, resulting in hemoconcentration of cells and plasma proteins, increasing the viscosity of the blood exiting the hemofilter. This filtration fraction (FF) can be calculated as the ratio of UF rate to plasma water, expressed as:

$$FF = \frac{Q_{UF}}{[Q_B \times (1 - Hct)]},$$

where Q_{UF} is the ultrafiltration rate, Q_B is the blood flow rate, and Hct is the patient's hematocrit. Optimally, the FF should be maintained <0.2 ; values exceeding 0.3 are associated with decreased hemofilter longevity [29]. FF may be minimized by increasing blood flow rate or by administering replacement fluid prior to the hemofilter. This latter strategy, while minimizing FF, decreases the concentration of solutes entering the hemofilter resulting in a lower solute concentration in the ultrafiltrate. The effective clearance (K_{eff}) is thus reduced by the ratio of the blood flow rate to the sum of the blood flow rate and prefilter replacement fluid (Q_R):

$$K_{eff} = Q_{UF} \times S \times \left[\frac{Q_B}{(Q_B + Q_R)} \right].$$

The characteristics of the hemofilter membrane determine the sieving coefficient for a given solute. Increasing the membrane surface area for a given membrane increases the hydraulic permeability and, hence, the maximal achievable UF rate. Thus, increasing membrane size allows for increased UF and increased clearance; however, if blood flow remains constant, this is at the expense of an increased UF rate. If UF is held constant, increasing the hemofilter surface area will have no effect on convective clearance.

15.5.2 Determinants of Diffusive Clearance

In conventional IHD, the dialysate flow rate usually exceeds blood flow rate and diffusive equilibration of solute between blood and dialysate is therefore incomplete. In contrast, during continuous hemodialysis, the blood flow rates utilized generally exceed dialysate flow allowing for near equilibration between blood and dialysate ($C_D/C_B \sim 1$) for

low-molecular-weight solutes ($<500-1500$ Da) such as urea [27]. If blood flow rate is held constant, as dialysate flow rate increases the degree of equilibration between blood and dialysate tends to fall [27]. For example, at a blood flow rate of 150 mL/min and a dialysate flow of 16.7 mL/min (1 L/h) equilibration of both urea and uric acid was essentially complete; when dialysate flow was increased to 41.7 mL/min (2.5 L/h) the fractional equilibration of urea decreased to 0.96 and fractional equilibration of uric acid decreased to 0.76 [27]. Despite this decline in equilibration between blood and dialysate, dialysate flow is the major determinant of low-molecular-weight solute clearance. Higher molecular weight solutes, such as β -2 microglobulin, are poorly cleared by diffusion and exhibit minimal dependence of dialysate flow [27]. It is likely that the clearance of these higher molecular weight solutes during continuous hemodialysis is the result of convective flux provided by unmeasured filtration/backfiltration within the hemodialyzer [25, 30]. This conjecture is supported by the observed lack of augmentation of clearance when diffusive therapy is added to convection in CVVHDF [28].

Unlike conventional IHD, under the operational conditions used during continuous hemodialysis and hemodiafiltration, diffusive clearance exhibits only minimal dependence on blood flow, so long as the blood flow rate is more than 3–5 times the dialysate flow rate [26, 31]. If solute equilibration between blood and dialysate is complete, increasing the surface area of the dialysis membrane will not augment clearance; however, if equilibration is incomplete, increasing the membrane surface may permit more complete equilibration and higher clearance rates.

15.5.3 Interaction Between Diffusive and Convective Clearance

Although in the discussion above, diffusion and convection were treated as independent processes, the overall clearance during hemodiafiltration may be less than the arithmetic sum of the clearances when dialysis (diffusive clearance) and hemofiltration (convective clearance) are performed separately [32, 33]. The magnitude of this interaction is related to the UF rate. When there is low convective flux minimal interaction is observed [26]; however, when higher UF rates are employed observed clearances during hemodiafiltration are less than predicted from the individual components of therapy [27, 30].

15.6 CRRT Extracorporeal Circuit

At a minimum, the extracorporeal circuit for CRRT consists of a vascular access which permits sufficient blood flow from and back to the patient, the hemofilter/hemodialyzer,

and pumps to regulate the UF rate, replacement fluid administration and/or dialysate inflow. In the AV therapies, the driving force for flow through the circuit is the pressure gradient between the mean arterial pressure and central venous pressure. In order to maximize flow, the resistance of the circuit needs to be minimized. The caliber of the catheters used for arterial and venous access need to be as large as the cannulated vessels will permit, their lengths should be relatively short to minimize resistance to flow and the overall extracorporeal circuit length should be kept as short as possible. Since the hemofilter/hemodialyzer represents the major site of resistance in the AV circuit, hemofilter/hemodialyzer configuration has a major effect on circuit function. Optimization of flow is provided by using hollow-fiber hemofilters/hemodialyzers with a large cross-sectional area and short axial length [34]. Increasing the diameter of the individual fibers decreases resistance, although at the expense of diminished effective surface area per unit volume of blood [34]. It has been suggested that the use of parallel plate hemodialyzers enhances diffusive clearance in CAVHD as compared to hollow-fiber dialyzers of equivalent surface area [35]. In the absence of a blood pump, pressure monitors and air detectors are generally not used in AV circuits.

Many of the considerations for optimization of AV circuits do not pertain to VV CRRT circuits. While the overall length of the VV CRRT circuit should not be excessive, there is not the same critical need for minimization and the tubing length should be sufficient to position the CRRT equipment without hindering patient access. Similarly, the need to minimize resistance to flow within the hemofilter/hemodialyzer does not pertain, and standard hemodialyzer configurations can be used. However, the use of a blood pump mandates the presence of an air detector with an integrated blood pump shutoff on the return line to minimize the risk of air embolization. The use of a pumped system also requires the use of pressure monitors to assess circuit patency, at a minimum on the inflow line, between the catheter and the blood pump, and on the return line, between the hemofilter/hemodialyzer and the catheter. Given the higher UF rates that can be achieved with pumped VV therapies, most VV therapies are now provided using dedicated equipment with fluid balancing capability to match dialysate and/or replacement fluid flow rates to effluent flow, thereby providing control of net fluid balance [36].

15.6.1 Vascular Access

Wide-bore arterial and venous catheters are the most common vascular access for AV CRRT [37]. Historically, AV (Scribner) shunts were also used; however, catheters generally provided higher blood flow rates, especially in the setting of hypotension [38]. Since it is readily cannulated and

provides high blood flow rates, the femoral artery is the most common site for arterial access; however, the brachial and axillary arteries in the upper limb and more distal vessels in both the upper and lower limbs may also be used. In order to optimize flow, the arterial catheter should be non-tapered with a single end-hole and should be of the largest diameter that the vessel can accommodate, preferably no smaller than 8 French [37]. Venous return can be through the femoral, subclavian, or internal jugular veins. The venous catheter should be relatively short, non-tapered with as large an internal diameter as feasible in order to minimize resistance to flow. The use of AV access, primarily the need for prolonged arterial cannulation with a large-bore catheter, is associated with more frequent and serious complications than seen with the VV therapies [39]. The major complications include bleeding, thromboembolism with distal limb ischemia, arterial aneurysms, or AV fistulas at the site of cannulation and infection. Caution must be used, particularly in patients with underlying peripheral vascular disease, to ensure that the arterial catheter does not compromise distal circulation. The catheter site and distal extremity must be monitored closely for evidence of infection or vascular injury while the catheter remains in place. The arterial catheter should be removed as soon as possible once treatment can be terminated. Anticoagulation should be reversed prior to removing the catheter and prolonged pressure applied after removal in order to assure adequate hemostasis.

Access for VV therapies may be obtained by placement of two separate single-lumen venous catheters, or more commonly, by the use of a standard double-lumen dialysis catheter. The KDIGO Clinical Practice Guideline for Acute Kidney Injury recommend the right jugular vein as the most optimal site for catheter placement, followed by the femoral veins, left internal jugular vein, and lastly the subclavian veins, with preference for placement on the dominant side [40]. The preference for the right internal jugular vein is based on lower rates of catheter dysfunction based on venous anatomy as compared to the left internal jugular vein [41]. There is a general preference for catheters above the waist as compared to catheters in the femoral veins due to increased risk of infection; however, in a randomized trial of 750 patients requiring dialysis for AKI the incidence of catheter-related blood stream infection was not higher in patients with femoral catheters [42]. Although subclavian catheters are associated with the lowest rate of infectious complications [43], they may result in the development of central venous stenosis which may complicate long-term vascular access in patients who do not recover kidney function and remain dialysis dependent [40, 44].

Ultrasound guidance during catheter insertion is recommended, particularly for insertion of catheters into the jugular veins [40, 45, 46]. A chest radiograph should be obtained to confirm position of internal jugular and subclavian cath-

eters prior to initial use [40]. Catheter position is important for optimization of catheter function. Depending on the specific catheter used, the catheter tip should either be at the junction of the superior vena cava and the right atrium or in the right atrium. When catheters are placed in the femoral position, the longest available catheter should be used so that the tip of the catheter is in the inferior vena cava. Aseptic techniques should be used during catheter insertion and during catheter care so as to minimize the risk of infection and use of the catheter should be restricted to dialytic therapies. Tunneled dialysis catheters can be used for CRRT, but are not recommended as the initial vascular access for most patients with AKI given an average duration of RRT of less than 2 weeks [40]. In patients with end-stage renal disease who require CRRT in the setting of critical illness, it is generally not recommended to use an existing AV fistula or graft for CRRT access given the risks of dislodgement or infiltration of rigid fistula needles during the course of care during prolonged therapy [47].

15.6.2 Hemofilter/Hemodialyzer

The membranes used for continuous hemofiltration must have a high hydraulic permeability in order to permit sufficient UF without necessitating excessive transmembrane pressure. This is in contrast to the membrane characteristics required for continuous hemodialysis, in which high UF rates are not required. The critical characteristic of a membrane used for continuous hemodialysis is its coefficient of diffusion, which is related to its composition, thickness, and surface area. While commercially available membranes are generally suitable for both hemofiltration and hemodialysis, some membranes that are optimized for convective therapy may provide inadequate diffusive clearance [31]. Although there has been controversy in the past regarding the effect of membrane biocompatibility on recovery of kidney function in AKI [48–53], the clinical relevance of this controversy is relatively limited given the wide availability of biocompatible synthetic membranes. The KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends the use of biocompatible synthetic membranes over cellulosic membranes for CRRT and other modalities of extracorporeal RRT in patients with AKI [40].

15.6.3 Pumps and Safety Monitors

Initial descriptions of pumped VV CRRT generally used a combination of a peristaltic blood pump to provide blood flow and volumetric infusion pumps to regulate flow of dialysate, replacement fluid, and effluent. Oftentimes, pressure monitors were not used and the only safety monitor was an

air detector on the venous return line. In addition, UF control was relatively poor as the linear peristaltic pumps used to regulate fluids and UF were not designed to operate against significant pressure gradients and were often prone to errors [54]. These improvised systems have been supplanted by integrated machines specifically designed for the performance of CRRT that are equipped with fluid-balancing technology to ensure delivery of the prescribed fluid management and integrated safety alarms, including air detectors and pressure monitors [36]. The equipment varies by manufacturer, with some machines capable of all modalities of VV CRRT while other machines can only provide continuous hemodialysis or hemodiafiltration but cannot provide hemodiafiltration.

15.7 Dialysate and Replacement Fluids

The use of dialysate and/or intravenous replacement solutions will depend on the modality of CRRT employed. In continuous hemofiltration, intravenous replacement fluids are administered to replace ultrafiltrate losses that exceed desired net fluid loss. In continuous hemodialysis, dialysate is perfused through the dialysate compartment of the hemodialyzer as in conventional IHD to provide diffusive clearance. In continuous hemodiafiltration, both dialysate and replacement fluids are utilized.

The compositions of dialysate and replacement fluid are similar; both should have an electrolyte composition similar to that of plasma, although often with lower potassium and higher buffer concentrations to permit correction of hyperkalemia and metabolic acidosis. The glucose concentration can vary; although some treatment protocols call for glucose-free solutions, most regimens use physiologic concentrations of approximately 100 mg/dL. Use of supraphysiologic glucose concentrations will result in glucose loading and may contribute to hyperglycemia. The concentration of calcium and magnesium are usually at physiologic to slightly supraphysiologic levels, with the exception that low or calcium-free solutions are often used in conjunction with citrate anticoagulation (see below). Most CRRT fluids contain no phosphate; although there is the potential for calcium phosphate deposition when phosphate is added to solutions containing calcium, stability has been demonstrated with phosphate concentrations of up to 1.2 mmol/L [55]. The addition of phosphate can prevent the development of hypophosphatemia and obviate the need for intravenous phosphate supplementation.

Various buffers have been used in fluids for CRRT including lactate, acetate, and bicarbonate. In the past, bicarbonate-buffered solutions could not be stored for prolonged periods of time and bicarbonate-buffered solutions for CRRT not were commercially available. The available solutions were buffered with either sodium lactate or sodium acetate, both of which are rapidly metabolized to bicarbonate. When lactate-

buffered solutions were used as dialysate or replacement fluid during CRRT, infusion rates could exceed metabolic clearance and modest elevations in blood lactate levels were common, although marked elevations were generally seen only in patients with underlying lactic acidosis or impaired hepatic metabolism [56–58]. Whether or not the modest elevations in lactate levels seen in most patients were associated with increased morbidity is not clear. Elevations in blood lactate and alterations in the pyruvate:lactate ratio are associated with increased protein catabolism and may contribute to myocardial depression. Several studies suggested that lactate-buffered fluids were less effective for controlling metabolic acidosis and were associated with increased hemodynamic instability as compared to bicarbonate-buffered fluids [59–61]. The KDIGO Clinical Practice Guideline for Acute Kidney Injury suggests that bicarbonate-buffered fluids be used rather than lactate-buffered solutions [40]. With the current ready availability of commercially prepared bicarbonate buffered fluids, these have generally replaced the use of acetate- and lactate-buffered CRRT solutions.

Replacement fluid must be sterile and pyrogen free; dialysate does not need to meet the same stringency of bacteriologic purity, as is also true for dialysate for conventional IHD, although commercially available dialysate for CRRT is both sterile and endotoxin free. Prior to the availability of commercially prepared fluids for CRRT, dialysate and replacement fluids were often compounded locally by pharmacy or by bedside nursing staff. Unfortunately, local preparation of solutions and the use of “custom” CRRT fluids has been associated with catastrophic compounding errors and should be discouraged [62, 63]. If fluids are compounded locally, they should be assayed prior to use to ensure that the electrolyte composition is correct.

15.8 Anticoagulation

Clotting of the extracorporeal circuit is the most common complication of CRRT and results in interruption of treatment and reduction in the delivered dose of therapy [64]. Multiple anticoagulation regimens have been used for prevention of clotting [65]. The optimal anticoagulation strategy will prevent clotting in the extracorporeal circuit while minimizing the risks of systemic bleeding; a particular concern in critically ill patients who often are in the immediate postoperative period or who have thrombocytopenia or underlying coagulopathy. Agents that have been utilized include unfractionated heparin, low-molecular-weight heparins, danaparoid, direct thrombin inhibitors such as hirudin and argatroban, nafamostat, and citrate [65].

The KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends using anticoagulation during CRRT in patients who do not have increased bleeding risk or impaired

coagulation and are not already receiving systemic anticoagulation [40]. Data from clinical trials suggest that 40–60% of patients undergoing CRRT receive therapy without anticoagulation [5, 6], the majority of whom are thrombocytopenic or coagulopathic. System patency in the absence of anticoagulation is optimized by maximizing the blood flow rate, ensuring that the FF is less than 30%, and minimizing interruptions in blood flow because of machine alarms or catheter malfunction. Restricted inflow of blood or increased resistance to blood return as the result of catheter malfunction is among the most common reasons for machine alarms and transient interruptions of treatment. Catheter malfunction may result from it being too short, malposition, kinking, or development of a fibrin sheath outside or an intraluminal clot within. If catheter malfunction is present, initiation or intensification of anticoagulation will provide minimal benefit and the catheter should be repositioned or replaced.

15.8.1 Heparin

Unfractionated heparin is the most commonly used anticoagulant for CRRT [5, 6, 66]. Heparin is infused into the afferent limb (pre-filter) of the extracorporeal circuit as close to the vascular access as possible. Heparin acts by potentiating the action of antithrombin III, resulting in inhibition of factors IIa (thrombin) and Xa. Inhibition of factor IIa prolongs the activated partial thromboplastin time (aPTT) allowing ready monitoring of heparin dosing. Multiple regimens for heparin infusion have been proposed. In general, an initial bolus of 10–30 IU/kg is given followed by an infusion of 5–10 IU/kg/h, with a target aPTT of 1.5–2.0 times the upper limit of normal. Regional anticoagulation with protamine reversal has been used in patients at high risk for bleeding [67, 68]. This approach is technically cumbersome and difficult to standardize, and the use of protamine may be associated with complications including hypotension, anaphylaxis, and thrombocytopenia [69]. Low-molecular-weight heparins have also been used for anticoagulation of extracorporeal dialysis circuits, primarily in the chronic IHD setting [70, 71]. Specific data in CRRT are limited; however, from several small trials low-molecular-weight heparins appear to be at least as efficacious and unfractionated heparin with similar profiles for bleeding risk [72, 73]. Because low-molecular-weight heparins predominantly inhibit factor Xa, they do not prolong aPTT and specific monitoring of anti-Xa activity is required (this may not be easily available). One limitation of low-molecular-weight heparins is that they accumulate in renal failure. The use of heparin is contraindicated in patients with heparin-induced thrombocytopenia and anticoagulation with argatroban (a direct thrombin inhibitor) or factor Xa inhibitors is recommended [40].

15.8.2 Citrate

Calcium is an obligate cofactor for coagulation; citrate chelates ionized calcium, thereby inhibiting the coagulation cascade. In regional citrate anticoagulation, citrate is infused into the afferent limb of the extracorporeal CRRT circuit as close to the vascular access as possible (ideally just where the afferent blood exits the catheter). Optimal anticoagulation is achieved when the ionized calcium level in the extracorporeal circuit is ≤ 0.35 mmol/L, measured post filter. Since the calcium–citrate complex is a small molecule, it will be readily cleared by either hemofiltration or hemodialysis. The remaining calcium–citrate complex that is returned to the body will be rapidly metabolized by the liver and skeletal muscle, generating three molecules of bicarbonate for each citrate molecule metabolized and releasing the bound calcium back into the circulation. In order to maintain calcium balance and prevent the development of systemic hypocalcemia, the calcium lost across the hemofilter/hemodialyzer membrane must be replaced with a systemic calcium infusion [65, 74]. The use of regional citrate anticoagulation requires close monitoring of the ionized calcium concentration in the return line of the extracorporeal circuit and systemically to ensure both adequate extracorporeal calcium chelation and maintenance of normal systemic ionized calcium concentration. Since citrate is metabolized to bicarbonate, acid-base status needs to be closely monitored to ensure that metabolic alkalosis does not develop. Depending on the specific citrate solution used, the buffer content of dialysate and/or replacement fluids may need to be reduced. In addition, many of the available citrate solutions have a high sodium content and can cause hyponatremia; if a hypertonic solution is used, the serum sodium concentration needs to be closely monitored and the sodium concentration of dialysate and/or replacement fluids decreased. Citrate accumulation may occur in patients with impaired citrate metabolism, such as patients with severe liver failure or lactic acidosis. In these patients, the use of citrate as an anticoagulant may be contraindicated as it may lead to citrate toxicity with severe systemic hypocalcemia [74–76]. Shock liver—which may coexist with ischemic AKI—is not of itself a contraindication to use of citrate. Multiple randomized trials have compared regional citrate anticoagulation to anticoagulation with either unfractionated or low-molecular-weight heparin in CRRT, finding increased circuit life and fewer bleeding events associated with citrate, with no increase in metabolic complications [77–84]. In one study, citrate anticoagulation was associated with improved renal recovery and hospital survival, although in this study, citrate anticoagulation was not associated with improved circuit patency [81]. Based on these data, the KDIGO Clinical Practice Guideline for Acute Kidney Injury suggests preferential use of citrate anticoagu-

lation over heparin for CRRT in patients who do not have a contraindication to citrate administration [40]. However, no citrate formulation is currently approved for use in anticoagulation of CRRT by the US Food and Drug Administration.

15.9 Comparison of Modalities of CRRT

The advantages of VV CRRT as compared to AV therapies have been previously discussed. The higher clearance of higher molecular weight solutes, including pro-inflammatory cytokines, has led to the hypothesis that continuous hemofiltration would have greater benefit, as compared to continuous hemodialysis, particularly in patients with sepsis-associated AKI [85, 86]. However, this hypothesis has not been borne out by clinical data. In a randomized controlled trial comparing high-volume to standard-volume CVVH in patients with sepsis-associated AKI, more intensive convective therapy was not associated with improved outcomes [87]. In a meta-analysis of 19 clinical trials, 16 of which utilized CRRT, no benefit was observed with convective therapies as compared to diffusive therapy [24]. Thus, based on current data, the modalities of CRRT should be considered equivalent and no recommendation can be made regarding use of continuous hemofiltration, hemodialysis, or hemodiafiltration.

15.10 Dosing of CRRT

As previously discussed, the clearance of urea and other low-molecular-weight molecules during CRRT is approximately equal to the total effluent flow rate, usually normalized to body weight. Several single-center randomized controlled trials suggested that CRRT delivered at effluent flow rates greater than 20 mL/kg/h were associated with improved patient outcomes as compared to CRRT at lower effluent flow rates [88, 89]; however, these results were not consistent across all trials [90, 91]. In two large, multi-center randomized controlled trials, higher doses of CRRT were not associated with improved clinical outcomes [5, 6]. In the Acute Renal Failure Trial Network (ATN) study, patients who received CVVHDF at an effluent flow rate of 35 mL/kg/h did not have improved survival at 60 days or recovery of kidney function as compared with patients who received CVVHDF at an effluent flow rate of 20 mL/kg/h [5]. Similarly, in the Randomized Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy study, which randomized 1508 patients to CVVHDF at an effluent flow of either 25 mL/kg/h or 40 mL/kg/h there was no difference in mortality at 90 days, ICU or hospital survival, or recovery of kidney function between groups [6]. Based on these results, the KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends delivering an effluent

volume of 20–25 mL/kg/h for CRRT in AKI [40]. Based on the observation that delivered therapy is often less than prescribed due to interruptions of therapy (leaving the ICU for imaging studies or therapeutic procedures or clotting of the extracorporeal circuit [64, 92]), the guideline suggests that this will usually require prescription of a higher effluent volume in order to achieve the target delivered dose [40]. Dosing of therapy must also take into account the individual patient's clinical status and the need to achieve adequate control of azotemia and electrolyte, acid-base, and volume status. In particular, higher doses of therapy may be needed in hypercatabolic patients in order to control hyperkalemia, metabolic acidosis, and azotemia.

15.11 Comparison of CRRT with Other Modalities of RRT

There has been substantial debate regarding the relative benefits of CRRT as compared to other modalities of RRT in patients with AKI. Individual randomized controlled trials comparing CRRT to conventional IHD have not demonstrated superiority of either modality of therapy [93–98]. Three meta-analyses pooling these data confirmed the absence of superiority for either intermittent or continuous therapy with regard to patient survival [99–101]. Observational studies have suggested that CRRT may be associated with improved recovery of kidney function as compared to conventional IHD. In a meta-analysis of 16 observational studies, conventional IHD was associated with an almost twofold increased risk of persistent dialysis dependence as compared to CRRT [102]. However, this benefit was not present in a pooled analysis of data from seven randomized controlled trials [102]. The reason for difference in the apparent benefit between the observational and randomized studies is most likely related to differences in the patients treated with intermittent as compared to continuous therapy in the observational trials, as evidenced by higher mortality rates in the CRRT cohorts in these studies. CRRT has been shown to be associated with a greater ability to achieve negative fluid balance than conventional IHD [103]. Given the absence of clear benefit of either modality, the KDIGO Clinical Practice Guideline for Acute Kidney Injury proposes that the modalities be considered complementary therapies in AKI patients and that the specific modality should be chosen based on patient characteristics as well as on available expertise and resources [40]. The guidelines do suggest, however, that CRRT be used preferentially over conventional intermittent therapy in hemodynamically unstable patients [40]. Data also suggest that CRRT may minimize the risk of compromised cerebral perfusion in patients with acute brain injury or other causes of increased intracranial pressure or cerebral edema [40, 104, 105]. There are only limited data comparing

CRRT to prolonged intermittent RRT; however, these data suggest similar metabolic and clinical outcomes [106, 107].

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